

# CUDA Practical Viva Questions

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## Question 1: What is CUDA?

### Answer:

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CUDA (Compute Unified Device Architecture) is a parallel computing platform and programming model developed by NVIDIA.

It allows programmers to use GPU resources for general-purpose parallel computing.

CUDA enables execution of thousands of threads simultaneously on the GPU.

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### Reasoning (Beginner Understanding):

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Normally, GPU is used for graphics rendering.

CUDA allows developers to use GPU power for:

- matrix multiplication,
- vector operations,
- AI/ML,
- image processing,
- simulations,
- and scientific computing.

GPU contains thousands of lightweight cores.

Therefore, large computations become much faster compared to CPU execution.

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## Question 2: Why is GPU faster than CPU for Parallel Computing?

## Answer:

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GPU is faster for parallel computing because it contains thousands of smaller cores designed for executing many operations simultaneously.

CPU contains fewer but more powerful cores optimized for sequential and control-heavy tasks.

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## Reasoning (Beginner Understanding):

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CPU is designed for:

- operating systems,
- logic handling,
- decision making,
- sequential programs.

GPU is designed for:

- repeated arithmetic operations,
- matrix calculations,
- parallel execution.

Suppose we must process 1 million elements.

CPU:

- processes fewer operations at a time.

GPU:

- distributes operations among thousands of threads.

This massively improves throughput.

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## Question 3: What is a CUDA Kernel?

### Answer:

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A CUDA Kernel is a special function executed on the GPU by many threads in parallel.

Kernels are launched using:

```
kernel<<<blocks, threads>>>();
```

cpp

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## Reasoning (Beginner Understanding):

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In normal C++:

- functions run on CPU.

In CUDA:

- kernels run on GPU.

When a kernel launches:

- thousands of GPU threads execute the same function simultaneously on different data.

Example:

- vector addition,
- matrix multiplication,
- image filtering.

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## Question 4: Explain the execution flow of a CUDA program.

### Answer:

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The execution flow of a CUDA program is:

1. Allocate memory on CPU (Host)
2. Initialize input data
3. Allocate memory on GPU (Device)
4. Copy data from Host to Device using `cudaMemcpy()`

5. Launch CUDA kernel
6. GPU performs parallel computation
7. Synchronize device using `cudaDeviceSynchronize()`
8. Copy result from Device to Host
9. Free allocated memory

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## Reasoning (Beginner Understanding):

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CPU and GPU have separate memory.

Therefore:

- data must first move from CPU memory to GPU memory.

Then:

- GPU executes the kernel in parallel.

Finally:

- result is copied back to CPU.

This complete workflow is one of the most important viva concepts.

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## Question 5: What is the difference between Host and Device in CUDA?

### Answer:

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| Host                       | Device                |
|----------------------------|-----------------------|
| CPU side                   | GPU side              |
| Executes normal C/C++ code | Executes CUDA kernels |
| Uses system RAM            | Uses GPU memory       |

| Host               | Device                        |
|--------------------|-------------------------------|
| Controls execution | Performs parallel computation |

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## Reasoning (Beginner Understanding):

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In CUDA terminology:

- CPU = Host
- GPU = Device

Host:

- prepares data,
- launches kernels,
- manages memory.

Device:

- performs parallel execution.

Both work together for GPU acceleration.

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## Question 6: What is thread indexing in CUDA?

### Answer:

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Thread indexing is the method used to uniquely identify each CUDA thread.

Common formula:

```
int i = blockIdx.x * blockDim.x + threadIdx.x;
```

cpp

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## Reasoning (Beginner Understanding):

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Every thread must know:

- which data element it should process.

Example:

- Thread 0 handles A[0]
- Thread 1 handles A[1]
- Thread 2 handles A[2]

Thread indexing provides unique IDs for parallel execution.

Without indexing:

- threads may process incorrect or duplicate data.
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## Question 7: What are Threads, Blocks, and Grids in CUDA?

### Answer:

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CUDA organizes execution hierarchically:

- Threads form a Block.
- Blocks form a Grid.

This structure helps GPU manage large-scale parallel execution.

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## Reasoning (Beginner Understanding):

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Suppose:

- one block contains 256 threads.
- 100 blocks exist.

Total threads:

```
256 × 100 = 25600 threads
```

```
text
```

Grid:

- collection of blocks.

Block:

- collection of threads.

Thread:

- performs actual computation.

This hierarchy improves GPU scheduling efficiency.

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## Question 8: Why do we use cudaMalloc()?

### Answer:

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`cudaMalloc()` is used to allocate memory on GPU device memory.

Syntax:

```
cudaMalloc((void**)&ptr, size);
```

```
cpp
```

### Reasoning (Beginner Understanding):

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CPU memory and GPU memory are separate.

Before GPU computation:

- memory must be allocated on GPU.

`cudaMalloc()` works similar to:

```
malloc()
```

```
cpp
```

but allocation happens on GPU memory instead of CPU RAM.

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## Question 9: What is cudaMemcpy() used for?

### Answer:

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`cudaMemcpy()` transfers data between CPU memory and GPU memory.

Syntax:

```
cudaMemcpy(destination, source, size, direction);
```

cpp

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### Reasoning (Beginner Understanding):

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GPU cannot directly access normal CPU variables.

Therefore data transfer is necessary.

Two common transfers:

```
Host → Device  
Device → Host
```

text

Example:

- input vectors copied to GPU.
- result copied back to CPU.

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## Question 10: Why is bounds checking important in CUDA kernels?

### Answer:

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Bounds checking prevents threads from accessing invalid memory locations when total threads exceed actual data size.

Example:

```
if(i < N)
```

cpp

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## Reasoning (Beginner Understanding):

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Suppose:

- vector size = 1000
- launched threads = 1024

Extra threads should not access memory beyond array size.

Without bounds checking:

- invalid memory access,
- segmentation faults,
- or incorrect outputs may occur.

Bounds checking ensures safe execution.

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## Question 11: What is `cudaDeviceSynchronize()`?

### Answer:

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`cudaDeviceSynchronize()` pauses CPU execution until all GPU kernel executions are completed.

It ensures proper synchronization between CPU and GPU.

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## Reasoning (Beginner Understanding):

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GPU execution is asynchronous.

That means:

- CPU may continue execution before GPU finishes computation.

If CPU tries to copy results too early:

- incorrect output may occur.

`cudaDeviceSynchronize()` ensures:

- GPU completes all tasks before moving forward.
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## Question 12: Why is Matrix Multiplication highly suitable for GPU execution?

### Answer:

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Matrix multiplication is highly suitable for GPU execution because many independent computations can run simultaneously.

Each output element of the result matrix can be computed independently by separate threads.

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## Reasoning (Beginner Understanding):

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Suppose:

$$C = A \times B$$

text

Each output element:

`C[row][col]`

text

is independent.

Therefore:

- one thread computes one output element.

Thousands of GPU threads can work simultaneously.

This matches GPU architecture perfectly.

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## Question 13: Why does Matrix Multiplication show more speedup than Vector Addition?

### Answer:

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Matrix multiplication shows greater speedup because it performs a much larger number of arithmetic operations compared to vector addition.

GPU resources become more effectively utilized.

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### Reasoning (Beginner Understanding):

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Vector addition performs:

```
A[i] + B[i]
```

text

only once per element.

Matrix multiplication performs:

- multiple multiplications,
- multiple additions,
- nested computations.

Therefore:

- GPU computation dominates,
- memory transfer overhead becomes smaller relative to computation.

This produces larger speedup.

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## Question 14: What is SIMT in CUDA?

### Answer:

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SIMT stands for Single Instruction Multiple Threads.

In CUDA, multiple threads execute the same instruction simultaneously on different data.

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### Reasoning (Beginner Understanding):

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Suppose vector addition is performed:

```
C[i] = A[i] + B[i]
```

text

All threads execute same instruction.

But:

- Thread 1 handles element 1
- Thread 2 handles element 2
- Thread 3 handles element 3

Same instruction + different data = SIMT.

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## Question 15: What is a Warp in CUDA?

### Answer:

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A warp is a group of 32 CUDA threads executed together by the GPU.

Warp is the basic scheduling unit in CUDA architecture.

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## Reasoning (Beginner Understanding):

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GPU does not execute threads individually.

Instead:

- threads are grouped into warps.

One warp contains:

```
32 threads
```

```
text
```

Efficient CUDA programming tries to keep all warp threads executing similar instructions.

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## Question 16: What is Warp Divergence?

### Answer:

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Warp divergence occurs when threads inside the same warp follow different execution paths due to conditional statements.

This reduces GPU efficiency.

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## Reasoning (Beginner Understanding):

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Suppose:

```
if(i % 2 == 0)
```

```
cpp
```

Some threads execute IF block. Other threads execute ELSE block.

GPU then executes branches separately.

This partially serializes execution and reduces parallel performance.

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## Question 17: What is Shared Memory in CUDA?

### Answer:

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Shared memory is a fast on-chip memory shared among threads within the same block. It provides much faster access compared to global memory.

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### Reasoning (Beginner Understanding):

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Global memory access is slower.

Shared memory acts like:

- a high-speed temporary workspace.

Threads inside same block can:

- share intermediate values,
- reduce repeated global memory access,
- improve performance.

Shared memory optimization is very important in CUDA.

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## Question 18: What is Global Memory in CUDA?

### Answer:

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Global memory is the main GPU memory accessible by all CUDA threads.

It has large storage capacity but higher access latency.

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## Reasoning (Beginner Understanding):

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Global memory stores:

- vectors,
- matrices,
- large datasets.

Advantages:

- accessible by all threads.
- large storage capacity.

Disadvantage:

- slower compared to shared memory.

Efficient CUDA programs minimize unnecessary global memory access.

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## Question 19: Why do we use 256 threads per block commonly?

### Answer:

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256 threads per block is commonly used because it provides efficient GPU utilization and aligns well with warp scheduling.

It is also a multiple of warp size (32).

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## Reasoning (Beginner Understanding):

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One warp contains:

32 threads

text

256 threads means:

This creates efficient GPU scheduling.

Too few threads:

- GPU underutilization.

Too many threads:

- resource limitations.

Therefore 256 is a practical balance.

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## Question 20: What is the purpose of kernel launch configuration?

### Answer:

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Kernel launch configuration specifies:

- number of blocks,
- number of threads per block,

used for GPU execution.

Example:

```
kernel<<<numBlocks, threadsPerBlock>>>();
```

cpp

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## Reasoning (Beginner Understanding):

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GPU execution requires work distribution.

Launch configuration determines:

- how many threads execute,
- how workload is divided,



- how GPU resources are utilized.

Incorrect configuration may:

- reduce performance,
  - waste GPU resources,
  - or cause execution errors.
- 

## Question 21: What is arithmetic intensity in CUDA?

### Answer:

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Arithmetic intensity refers to the amount of computation performed compared to memory access.

It is calculated as:

$$\text{Arithmetic Operations} / \text{Memory Operations}$$

text

Higher arithmetic intensity usually leads to better GPU performance.

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### Reasoning (Beginner Understanding):

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Suppose:

- a program performs many calculations after loading data.

Then:

- GPU computation becomes efficient.

Matrix multiplication has high arithmetic intensity because:

- one memory load participates in many computations.

Vector addition has low arithmetic intensity because:

- very little computation occurs per memory access.

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## Question 22: Why is memory transfer overhead important in CUDA?

### Answer:

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Memory transfer overhead is important because data transfer between CPU and GPU consumes significant time.

Sometimes transfer time becomes larger than actual GPU computation time.

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### Reasoning (Beginner Understanding):

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Before GPU computation:

- data moves from CPU RAM to GPU memory.

After computation:

- results move back to CPU.

These transfers take time.

For small computations:

- transfer overhead may dominate.

That is why GPU acceleration is more beneficial for large computational workloads.

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## Question 23: What is the purpose of **global** keyword in CUDA?

### Answer:

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`__global__` is used to declare a CUDA kernel function that executes on the GPU and can be called from the CPU.

Example:

```
__global__ void vectorAdd()
```

cpp

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## Reasoning (Beginner Understanding):

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CUDA requires special keywords to specify execution location.

`__global__` means:

- function runs on GPU,
- function is launched from CPU.

Without `__global__`:

- CUDA kernel cannot execute on GPU.

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## Question 24: What is the purpose of device keyword in CUDA?

### Answer:

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`__device__` declares a function or variable that exists and executes only on the GPU.

It cannot be directly called from CPU code.

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## Reasoning (Beginner Understanding):

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`__device__` functions are helper functions used internally by GPU kernels.

Example:

- mathematical helper computations,
- intermediate GPU operations.

CPU cannot directly access these functions.

They exist only inside GPU execution environment.

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## Question 25: What is the purpose of **host** keyword in CUDA?

### Answer:

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`__host__` specifies that a function executes on the CPU.

Normal C/C++ functions are host functions by default.

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### Reasoning (Beginner Understanding):

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CUDA programs contain:

- CPU code,
- GPU code.

`__host__` explicitly indicates:

- function runs on CPU.

Usually:

- memory management,
- input handling,
- kernel launching,

occur on Host side.

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## Question 26: What is occupancy in CUDA?

## Answer:

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Occupancy refers to how effectively GPU resources are utilized by active warps on a Streaming Multiprocessor (SM).

Higher occupancy can improve GPU performance.

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## Reasoning (Beginner Understanding):

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GPU tries to keep many threads active simultaneously.

If too few threads are active:

- GPU cores remain idle.

Higher occupancy means:

- better utilization of GPU hardware,
- improved parallel efficiency.

However:

- maximum occupancy does not always guarantee maximum performance.
- 

## Question 27: What is memory coalescing in CUDA?

### Answer:

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Memory coalescing is an optimization technique where nearby threads access nearby memory locations efficiently.

This reduces memory transactions and improves performance.

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## Reasoning (Beginner Understanding):

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Suppose:

- consecutive threads access consecutive memory addresses.

GPU combines these accesses into fewer memory operations.

This improves:

- memory bandwidth utilization,
- execution speed.

Poor memory access patterns reduce CUDA performance.

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## Question 28: Why is Shared Memory faster than Global Memory?

### Answer:

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Shared memory is faster because it is located on-chip near GPU cores, while global memory is located off-chip with higher latency.

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### Reasoning (Beginner Understanding):

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Shared memory acts like:

- a small high-speed workspace.

Global memory behaves like:

- large but slower storage.

Accessing shared memory:

- takes fewer clock cycles.

Therefore:

- CUDA optimization often focuses on maximizing shared memory usage.
-

## Question 29: What is tiled matrix multiplication in CUDA?

### Answer:

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Tiled matrix multiplication is an optimization technique where small blocks (tiles) of matrices are loaded into shared memory before computation.

This reduces repeated global memory access.

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### Reasoning (Beginner Understanding):

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Normal matrix multiplication repeatedly accesses global memory.

This becomes slow.

Tiling works by:

- loading small matrix portions into shared memory,
- reusing them multiple times.

Benefits:

- faster computation,
  - reduced memory latency,
  - improved GPU utilization.
- 

## Question 30: Why is CUDA important in AI and Machine Learning?

### Answer:

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CUDA is important in AI and Machine Learning because deep learning involves massive matrix and tensor operations that can be executed efficiently on GPUs.

CUDA accelerates training and inference significantly.

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## Reasoning (Beginner Understanding):

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Neural networks perform:

- matrix multiplication,
- convolution operations,
- tensor calculations,
- gradient updates.

These operations are highly parallel.

GPU executes thousands of these operations simultaneously.

Without CUDA acceleration:

- deep learning training would become extremely slow.
- 

# Question 31: What is the difference between Sequential and Parallel Execution?

## Answer:

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| Sequential Execution            | Parallel Execution                    |
|---------------------------------|---------------------------------------|
| Tasks execute one after another | Multiple tasks execute simultaneously |
| Uses fewer processing resources | Uses multiple threads/cores           |
| Slower for large computations   | Faster for large computations         |
| Easier to implement             | Requires synchronization              |

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## Reasoning (Beginner Understanding):

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In sequential execution:

```
Task1 → Task2 → Task3
```

text

In parallel execution:

```
Task1  
Task2 → Executed Together  
Task3
```

text

CUDA uses parallel execution to improve performance significantly.

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## Question 32: Why do we use GPU for Matrix Multiplication?

### Answer:

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GPU is used for matrix multiplication because matrix operations contain massive independent computations that can execute simultaneously.

GPU architecture is optimized for such parallel workloads.

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## Reasoning (Beginner Understanding):

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Each matrix element calculation is independent.

Example:

```
C[0][0]  
C[0][1]  
C[1][0]
```

text

All can be computed simultaneously.

GPU distributes these calculations among thousands of threads.

This produces major speed improvements.

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## Question 33: Why do we compare CPU and GPU execution times?

### Answer:

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CPU and GPU execution times are compared to analyze performance improvement achieved through parallel execution.

This helps measure:

- speedup,
  - efficiency,
  - scalability.
- 

### Reasoning (Beginner Understanding):

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Suppose:

- CPU takes 10 seconds.
- GPU takes 2 seconds.

Then:

$$\text{Speedup} = 10 / 2 = 5x$$

text

This comparison proves:

- GPU acceleration benefits.

Performance analysis is an important part of HPC.

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## Question 34: Why does GPU not always outperform CPU?

### Answer:

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GPU does not always outperform CPU because:

- memory transfer overhead exists,
  - some problems are too small,
  - sequential dependencies reduce parallelism.
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### Reasoning (Beginner Understanding):

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GPU acceleration becomes beneficial when:

- large-scale parallel computation exists.

For small problems:

- transfer overhead may dominate.

CPU is better for:

- control-heavy tasks,
- sequential logic,
- branching-intensive programs.

Therefore:

- GPU is not universally faster.
- 

## Question 35: What is latency in GPU computing?

### Answer:

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Latency is the delay between requesting an operation and receiving the result.

GPU memory accesses often have higher latency compared to computation speed.

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## Reasoning (Beginner Understanding):

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Suppose:

- GPU requests data from memory.

Time required to receive data:

- is latency.

CUDA optimization tries to:

- hide latency,
- keep GPU busy using many active threads.

High parallelism helps reduce visible latency impact.

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## Question 36: What is throughput in GPU computing?

### Answer:

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Throughput refers to the amount of work completed per unit time.

GPU architecture is optimized for very high throughput.

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## Reasoning (Beginner Understanding):

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CPU focuses on:

- fast completion of single tasks.

GPU focuses on:

- completing massive numbers of operations together.

Example:

- processing millions of vector elements simultaneously.

This makes GPUs highly efficient for scientific and AI workloads.

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## Question 37: What is the purpose of synchronization in CUDA?

### Answer:

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Synchronization ensures correct coordination between threads and prevents incorrect data access during parallel execution.

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### Reasoning (Beginner Understanding):

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Suppose:

- one thread writes data,
- another thread reads same data immediately.

Without synchronization:

- incorrect or incomplete values may be read.

Synchronization ensures:

- all required computations complete before dependent operations begin.
- 

## Question 38: What are the limitations of Parallel Computing?

## Answer:

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Limitations of Parallel Computing include:

1. Synchronization overhead
  2. Communication overhead
  3. Race conditions
  4. Complex debugging
  5. Sequential dependencies
  6. Hardware limitations
- 

## Reasoning (Beginner Understanding):

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Not all problems can be perfectly parallelized.

Some tasks depend on previous computations.

Additional problems include:

- thread coordination,
- memory conflicts,
- inefficient workload distribution.

Therefore parallel programming is more complex than sequential programming.

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## Question 39: What is Speedup in CUDA?

### Answer:

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Speedup measures how much faster GPU execution is compared to CPU execution.

Formula:

$$\text{Speedup} = \text{CPU Time} / \text{GPU Time}$$

text

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## Reasoning (Beginner Understanding):

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Suppose:

- CPU execution = 20 seconds
- GPU execution = 4 seconds

Then:

$$\text{Speedup} = 20 / 4 = 5$$

text

Meaning:

- GPU execution is 5 times faster.

Higher speedup indicates better GPU acceleration.

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## Question 40: Why is Parallel Computing important in modern applications?

### Answer:

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Parallel Computing is important because modern applications require extremely high computational power and fast processing.

It enables efficient execution of:

- AI/ML,
  - simulations,
  - gaming,
  - scientific computing,
  - big data analytics,
  - and cloud computing.
- 

## Reasoning (Beginner Understanding):

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Modern applications process:

- massive datasets,
- complex algorithms,
- real-time computations.

Sequential execution becomes too slow.

Parallel computing solves this problem by:

- distributing workload across multiple processors and threads.

This improves:

- speed,
- scalability,
- and overall system performance.

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## Question 41: What is the difference between Vector Addition and Matrix Multiplication in CUDA?

Answer:

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| Vector Addition               | Matrix Multiplication        |
|-------------------------------|------------------------------|
| Simple element-wise operation | Complex arithmetic operation |
| Low arithmetic intensity      | High arithmetic intensity    |
| Fewer computations            | Large number of computations |
| Smaller GPU speedup           | Higher GPU speedup           |
| Memory-bound                  | Compute-bound                |

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## Reasoning (Beginner Understanding):

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Vector addition performs:

```
C[i] = A[i] + B[i]
```

text

only once per element.

Matrix multiplication performs:

- multiple multiplications,
- multiple additions,
- nested computations.

Therefore:

- GPU utilization becomes much higher in matrix multiplication.

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## Question 42: What is Memory Bandwidth in GPU computing?

### Answer:

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Memory bandwidth refers to the amount of data that can be transferred between GPU memory and GPU processors per second.

Higher memory bandwidth improves GPU performance.

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## Reasoning (Beginner Understanding):

Suppose GPU cores require data very quickly.

If memory transfer is slow:

- GPU cores remain idle.

High memory bandwidth ensures:

- faster data movement,
  - better GPU utilization,
  - improved execution speed.
- 

## Question 43: Why is CUDA considered a parallel programming model?

### Answer:

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CUDA is considered a parallel programming model because it allows thousands of GPU threads to execute computations simultaneously.

It distributes large workloads across GPU cores.

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### Reasoning (Beginner Understanding):

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Instead of one processor performing all work:

- CUDA divides work among many GPU threads.

Example:

- each thread computes one vector element.
- each thread computes one matrix element.

This simultaneous execution is the foundation of parallel programming.

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## Question 44: What is a Streaming Multiprocessor (SM) in CUDA?

### Answer:

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A Streaming Multiprocessor (SM) is the main processing unit inside a CUDA GPU responsible for executing warps and managing threads.

Each SM contains:

- CUDA cores,
  - registers,
  - shared memory,
  - schedulers.
- 

## Reasoning (Beginner Understanding):

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GPU is divided into multiple SMs.

Each SM handles:

- thread scheduling,
- warp execution,
- memory operations.

More SMs generally provide:

- higher parallel processing capability.
- 

## Question 45: Why is parallel debugging difficult?

### Answer:

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Parallel debugging is difficult because multiple threads execute simultaneously and interact with shared resources.

Errors may occur unpredictably.

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## Reasoning (Beginner Understanding):

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Sequential programs execute in fixed order.

Parallel programs:

- execute threads concurrently.
- execution order may vary.

Problems include:

- race conditions,
- deadlocks,
- synchronization bugs,
- nondeterministic outputs.

These issues are harder to trace and reproduce.

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## Question 46: What is a race condition in GPU programming?

### Answer:

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A race condition occurs when multiple GPU threads access and modify shared data simultaneously without proper synchronization.

This can produce incorrect or unpredictable results.

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## Reasoning (Beginner Understanding):

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Suppose two threads execute:

```
sum += value;
```

cpp

simultaneously.

Both threads may read same old value.

One update may overwrite another.

Result:

- incorrect output.

Race conditions are prevented using:

- synchronization,
  - atomic operations,
  - careful memory handling.
- 

## **Question 47: What is the role of CUDA in High Performance Computing (HPC)?**

### **Answer:**

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CUDA plays a major role in HPC by accelerating computationally intensive workloads using GPU parallelism.

It improves performance for:

- simulations,
  - AI/ML,
  - scientific computing,
  - matrix operations,
  - image processing,
  - and big data analytics.
- 

### **Reasoning (Beginner Understanding):**

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HPC problems involve:

- huge datasets,
- massive computations,
- long execution times.

CUDA distributes these computations across thousands of GPU cores.

This significantly reduces execution time and increases throughput.

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## **Question 48: Why is Matrix Multiplication important in AI and Deep Learning?**

### **Answer:**

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Matrix multiplication is fundamental in AI and Deep Learning because neural networks perform repeated matrix and tensor operations during training and inference.

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### **Reasoning (Beginner Understanding):**

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Neural networks contain:

- inputs,
- weights,
- activations.

Forward propagation and backpropagation involve:

- large matrix multiplications.

GPU acceleration through CUDA makes deep learning practical and efficient.

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## **Question 49: What optimizations can improve CUDA program performance?**

### **Answer:**

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CUDA performance can be improved using:

1. Shared memory optimization
  2. Memory coalescing
  3. Tiled matrix multiplication
  4. Efficient thread/block configuration
  5. Reducing memory transfers
  6. Increasing occupancy
  7. Minimizing warp divergence
- 

## Reasoning (Beginner Understanding):

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GPU performance depends on:

- memory efficiency,
- thread efficiency,
- workload distribution.

Optimizations try to:

- reduce delays,
- improve hardware utilization,
- maximize parallel execution.

Efficient CUDA programs focus heavily on optimization techniques.

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## Question 50: What are the main advantages of CUDA programming?

### Answer:

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Main advantages of CUDA programming are:

1. Massive parallel execution
2. Faster computation
3. GPU acceleration

4. High throughput
  5. Better performance for AI/ML
  6. Efficient scientific computation
  7. Scalability for large workloads
- 

## **Reasoning (Beginner Understanding):**

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CUDA allows programmers to use GPU hardware for general-purpose computing.

Benefits include:

- reduced execution time,
- handling large computations efficiently,
- acceleration of AI and scientific applications.

CUDA is one of the most important technologies in modern High Performance Computing.

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